THE DIAGONAL OF THE COMPLEXES STACK IS REPRESENTABLE BY ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file quot.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Situation situation-complexes (canonical tag 0DLE). cited at statement line 5427. Standing situation.
- Lemma algebraic-lemma-representable-diagonal (canonical tag 045G). cited at proof lines 5436-5437. Goal reformulation: Isom over T is an algebraic space.
- Lemma lemma-complexes (canonical tag 0DLD). cited at proof lines 5450-5451. E, E' satisfy conditions (1), (2).
- Lemma lemma-complexes-open-neg-exts-vanishing (canonical tag 0DLC). cited at proof lines 5462-5463 and 5476-5477. Opens W, W' and affine fp representability of the SheafHom functors.
- Proposition proposition-isom (canonical tag 08K9). cited at proof lines 5482-5483. The Isom-as-fibre-product presentation.

Source reference. Original definition 0DLE: sources/stacks-chow/tags.txt, source lines 17044–17044. Cited at statement line 5427: the lemma is phrased in this situation; load-bearing.

Source reference. Original lemma 045G: sources/stacks-chow/tags.txt, source lines 5435–5435. Cited at proof lines 5436-5437: the goal reformulation; load-bearing.

Source reference. Original lemma 0DLD: sources/stacks-chow/tags.txt, source lines 17043–17043. Cited at proof lines 5450-5451: E, E' satisfy conditions (1), (2); load-bearing.

Source reference. Original lemma 0DLC: sources/stacks-chow/tags.txt, source lines 17042–17042. Cited at proof lines 5462-5463 and 5476-5477 (twice): the opens W, W' and the representability of H = SheafHom(E, E') and friends; load-bearing twice.

Source reference. Original proposition 08K9: sources/stacks-chow/tags.txt, source lines 10877–10877. Cited at proof lines 5482-5483: repeating its argument gives Isom(E, E') = (H' x_B H) x_{c, I x_B I', sigma} B; load-bearing.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *In Situation 0DLE. Denote $\mathcal{X} = \mathit{Complexes}_{X/B}$. Then $\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces.*

Proof. Consider two objects $x = (T, g, E)$ and $y = (T, g', E')$ of $\mathcal{X}$ over a scheme T. We have to show that $\mathit{Isom}_{\mathcal{X}}(x, y)$ is an algebraic space over T, see Algebraic Stacks, Lemma 045G. If for $h : T' \to T$ the restrictions $x|_{T'}$ and $y|_{T'}$ are isomorphic in the fibre category $\mathcal{X}_{T'}$, then $g \circ h = g' \circ h$. Hence there is a transformation of presheaves

$$\mathit{Isom}_{\mathcal{X}}(x, y) \longrightarrow \text{Equalizer}(g, g')$$

Since the diagonal of B is representable (by schemes) this equalizer is a scheme. Thus we may replace T by this equalizer and E and E' by their pullbacks. Thus we may assume $g = g'$.

Assume $g = g'$. After replacing B by T and X by X_T we arrive at the following problem. Given $E, E' \in D(\mathcal{O}_X)$ satisfying conditions (1), (2) of Lemma 0DLD we have to show that $\mathit{Isom}(E, E')$ is an algebraic space. Here $\mathit{Isom}(E, E')$ is the functor

$$(\mathit{Sch}/B)^{opp} \to \mathit{Sets}, \quad T \mapsto \{\varphi : E_T \to E'_T \text{ isomorphism in } D(\mathcal{O}_{X_T})\}$$

where E_T and E'_T are the derived pullbacks of E and E' to X_T. Now, let $W \subset B$, resp. $W' \subset B$ be the open subspace of B associated to E, E', resp. to E', E by Lemma 0DLC. Clearly, if there exists an isomorphism $E_T \to E'_T$ as in the definition of $\mathit{Isom}(E, E')$, then we see that $T \to B$ factors into both W and W' (because we have condition (1) for E and E' and we'll obviously have $E_t \cong E'_t$ so no nonzero maps $E_t[i] \to E_t$ or $E'_t[i] \to E_t$ over the fibre X_t for $i > 0$. Thus we may replace B by the open $W \cap W'$. In this case the functor $H = \mathcal{H}om(E, E')$

$$(\mathit{Sch}/B)^{opp} \to \mathit{Sets}, \quad T \mapsto \text{Hom}_{\mathcal{O}_{X_T}}(E_T, E'_T)$$

is an algebraic space affine and of finite presentation over B by Lemma 0DLC. The same is true for $H' = \mathcal{H}om(E', E)$, $I = \mathcal{H}om(E, E)$, and $I' = \mathcal{H}om(E', E')$. Therefore we can repeat the argument of the proof of Proposition 08K9 to see that

$$\mathit{Isom}(E, E') = (H' \times_B H) \times_{c, I \times_B I', \sigma} B$$

for some morphisms c and σ. Thus $\mathit{Isom}(E, E')$ is an algebraic space. □